

Discovery of the surface polarity gradient on iridescent *Morpho* butterfly scales reveals a mechanism of their selective vapor response

Radislav A. Potyrailo^{a,1,2}, Timothy A. Starkey^{b,1}, Peter Vukusic^b, Helen Ghiradella^c, Milana Vasudev^d, Timothy Bunning^d, Rajesh R. Naik^d, Zhexiong Tang^a, Michael Larsen^a, Tao Deng^{a,e}, Sheng Zhong^a, Manuel Palacios^a, James C. Grande^a, Gilad Zorn^a, Gregory Goddard^a, and Sergey Zalubovsky^a

^aGeneral Electric Global Research Center, Niskayuna, NY 12309; ^bSchool of Physics, University of Exeter, Exeter EX4 4QL, United Kingdom; ^cDepartment of Biological Sciences, University at Albany, State University of New York, Albany, NY 12222; ^dMaterials and Manufacturing Directorate, Air Force Research Laboratory, Wright-Patterson Air Force Base, OH 45433; and ^eState Key Laboratory of Metal Matrix Composites, School of Materials Science and Engineering, Shanghai Jiao Tong University, Shanghai 200240, People's Republic of China

Edited by David A. Weitz, Harvard University, Cambridge, MA, and approved August 6, 2013 (received for review June 16, 2013)

For almost a century, the iridescence of tropical *Morpho* butterfly scales has been known to originate from 3D vertical ridge structures of stacked periodic layers of cuticle separated by air gaps. Here we describe a biological pattern of surface functionality that we have found in these photonic structures. This pattern is a gradient of surface polarity of the ridge structures that runs from their polar tops to their less-polar bottoms. This finding shows a biological pattern design that could stimulate numerous technological applications ranging from photonic security tags to self-cleaning surfaces, gas separators, protective clothing, sensors, and many others. As an important first step, this biomaterial property and our knowledge of its basis has allowed us to unveil a general mechanism of selective vapor response observed in the photonic *Morpho* nanostructures. This mechanism of selective vapor response brings a multivariable perspective for sensing, where selectivity is achieved within a single chemically graded nanostructured sensing unit, rather than from an array of separate sensors.

multivariable sensing | spectral analysis | surface characterization | transmission electron microscopy | image analysis

Structural colors in tropical butterflies have been of scientific interest for almost a century (1, 2), with many optical studies examining *Morpho* species (3, 4). The iridescence exhibited by scales of *Morpho* butterflies originates from 3D vertical ridge structures of stacked periodic layers of cuticle separated by air gaps (5).

In this study, we have found a biological pattern of surface functionality within these ridge structures of *Morpho* scales. This pattern is a naturally formed gradient of surface polarity extending from the polar tops of ridges to their less-polar bottoms. We validated the existence of this gradient by applying a suite of complementary techniques to assess the chemistry of the ridges on the nanoscale. At first, we mapped the spatial polarity distribution of individual ridges by staining scales with polarity-sensitive dyes and performing spatially resolved optical characterization experiments. We further monitored the change of spatial polarity distribution within individual ridges before and after a control experiment aimed to reduce this initial polarity gradient. Finally, we performed detailed experiments of exposing *Morpho* scales to vapors of different polarity, observed the optical spectral responses, and compared these experimental effects with responses from simulated *Morpho* nanostructures with either a uniform or a gradient surface polarity. Results of these complementary techniques provided strong evidence of the existence of the naturally formed gradient of surface polarity on the ridges of *Morpho* scales.

Although this surface polarity gradient may not be essential for butterfly survival, but rather is a by-product of the process of scale development, this perspective on biological pattern design

can offer opportunities for a variety of technological applications. Examples of such applications include photonic security tags, self-cleaning surfaces, gas separators, protective clothing, sensors, and many others. As a first application, the knowledge of the gradient of surface polarity of the ridges allowed us to unveil a general mechanism of selective vapor response in photonic *Morpho* nanostructures. According to this mechanism, vapors of different polarity are preferentially adsorbed onto the respective regions of the ridges. This preferential adsorption is expressed in the corresponding spectral regions of the reflectance spectra. Our previous work on the vapor response of *Morpho* butterfly scales (6) did not provide a mechanism for the observed unusually selective vapor response.

In vapor sensing, the quest for selective sensors started in the 1950s with studies of metal oxides (7). Following initial observations of vapor responses of different materials, competing requirements between selectivity and reversibility of sensors were discovered, explaining poor selectivity of individual sensors (8). The idea of combining individual sensors into arrays (9) became an accepted compromise to improve selectivity. At present, development of selectivity-tunable yet simplified sensor systems

Significance

Morpho butterflies are a brilliant spectacle of nature's capability for photonic engineering. Their conspicuous appearance arises from the interference and diffraction of light within tree-like nanostructures on their scales. Scientific lessons learned from these butterflies have already inspired designs of new displays, fabrics, and cosmetics. This study reports a vertical surface polarity gradient in these tree-like structures. This biological pattern design may be applied to numerous technological applications ranging from security tags to self-cleaning surfaces, gas separators, protective clothing, and sensors. Here it has allowed us to unveil a general mechanism of selective vapor response in photonic *Morpho* nanostructures and to demonstrate attractive opportunities for chemically graded sensing units for high-performance sensing.

Author contributions: R.A.P., P.V., H.G., T.B., R.R.N., and S. Zalubovsky designed research; R.A.P., T.A.S., M.V., Z.T., M.L., T.D., and S. Zhong performed research; R.A.P., P.V., H.G., M.V., T.B., R.R.N., Z.T., M.P., J.C.G., G.Z., G.G., and S. Zalubovsky contributed new reagents/analytic tools; R.A.P., T.A.S., M.V., M.L., T.D., S. Zhong, M.P., J.C.G., G.Z., and G.G. analyzed data; and R.A.P. and T.A.S. wrote the paper.

The authors declare no conflict of interest.

This article is a PNAS Direct Submission.

¹R.A.P. and T.A.S. contributed equally to this work.

²To whom correspondence should be addressed. E-mail: potyrailo@crd.ge.com.

This article contains supporting information online at www.pnas.org/lookup/suppl/doi:10.1073/pnas.1311196110/-DCSupplemental.

methanol, ethanol, propanol, butanol, and water vapors and analyzed spectral responses using PCA (Fig. 3D). To maximize response reproducibility, we controlled vapor exposures of the *Morpho* samples (Methods and Fig. S3) and achieved measurement precision of <10% relative SD. Vapors were discriminated by the *Morpho* scales with the response order related to the polarities of water, methanol, and ethanol followed by propanol and butanol, which were not as clearly resolved. These responses to different vapors formed individual response directions in the PCA scores plot.

In a further demonstration, the *Morpho* scales were challenged with closely related nonpolar vapors with similar refractive indices and different polarities (three isomers of dichloroethylene). The results confirmed the polarity-dependent, rather than refractive-index-dependent, response (Fig. S4). Finally, detection of vapors of different refractive indices and different polarities before and after a mild oxygen plasma treatment of the scales revealed that the plasma treatment altered the initial polarity-dependent response to a refractive-index-dependent response (Fig. S5).

Such selectivity of response to different vapors of *Morpho* scales is a significant advancement over existing nanofabricated photonic sensors that use porous silicon (27), opal structures (24), inverse opal structures (28), and other chemically uniform dielectric nanostructures based on the constructive interference of light that results in the Bragg diffraction. These known sensors exhibit vapor response as the wavelength shift $\Delta\lambda$ of the Bragg wavelength in the reflectance spectrum in proportion to the product of the vapor refractive index n and the adsorbed layer thickness d : $\Delta\lambda \sim n \times d$ (27). These sensors with individual Bragg wavelength stop-bands cannot discriminate vapors (29, 30) because chemically uniform nanostructures facilitate a uniform adsorption of vapors onto the nanostructure and are not expected to form individual response directions in the PCA scores plot.

Thus, we compared the multivapor response profiles of *Morpho* scales and conventional sensors based on Bragg diffraction. As an example, we fabricated a classic chemically uniform porous silicon structure, exposed it to the same five vapors, and compared the response profiles (Fig. S6). We used porous silicon as a spectral control because of its high vapor sensitivity and well-understood vapor response mechanism (23, 27, 29, 31). As predicted from the Bragg wavelength shift relationship, responses of this conventional sensing material to the five vapors have lined up only as $n \times d$ and did not form individual response directions in the PCA scores plot (Fig. S7A). The combined PCA results for the *Morpho* scales and porous silicon (Fig. S7B) demonstrated the high level of vapor discrimination by the *Morpho* scales and no vapor discrimination by this conventional sensing material. This comparison of the spectral diversity of vapor responses of the *Morpho* scales and a typical gradient-free photonic nanostructured sensing material further validated the role of the found surface polarity gradient on *Morpho* scales in their vapor-response selectivity.

Mechanism for the Selective Vapor Response of *Morpho* Scales Based on Polarity Gradient of the Ridge Structure. After the experimental validation of the spectral diversity of responses of the *Morpho* scales to vapors of different polarity, we further computed the spectral responses upon two scenarios of coverage of the photonic *Morpho* nanostructure with different adsorbed vapors. These vapor-coverage scenarios included a uniform and a gradient coverage. The refractive index of the structure, $n = 1.56$, and its extinction coefficient, $k = 0.06$, were assumed to be the same as determined for the natural *Morpho* nanostructures (5) and not affected by adsorption of vapors. For these simulations we used vapors of three model solvents with $n = 1.3$, 1.4, and 1.5, their three vapor concentrations being related to condensed liquid layers of thickness $d = 5$, 10, and 15 nm formed on the nanostructure. These values of condensed liquid layers were selected

on the basis of the reliably achievable spectral resolution of optical simulations and thickness of liquid layers of condensed vapors on porous films reported in the literature (20, 32).

The uniform coverage of the photonic structure with different adsorbed vapors was simulated as condensed liquid layers conformally covering the whole nanostructure. The gradient distribution of vapor adsorption was approximated as three regions covering only the top, middle, and bottom regions of the ridges. Although simulations were only performed for three vapors and three spatial regions, this approach was sufficient to explore the premise that the vapor selectivity originates from a polarity gradient effect. Complete sets of reflectance spectra $R(\lambda)$ of uniform and gradient vapor adsorption onto the *Morpho* nanostructure are available in Fig. S8, with representative examples illustrated in Fig. 4A and C. All simulated spectra were processed using PCA; the scores plots are presented in Fig. 4B and D.

The uniform coverage by different vapors produced a response related to the optical thickness $n \times d$ of the conformal film. The reflectance spectra (Fig. 4A and Fig. S8A) were correlated to the $n \times d$ product, being visualized as a single response line in the PCA scores plot (Fig. 4B). This vapor response was similar to that of conventional sensing materials based on Bragg diffraction, including porous silicon as measured in this study for comparison (Fig. S6). Thus, the uniform coverage of the *Morpho* nanostructure with three different vapors did not explain the experimentally observed response to different vapors that formed individual response directions in the PCA scores plot (6).

In contrast, simulations of gradient coverage by different vapors produced reflectance spectra illustrated in Fig. 4C and

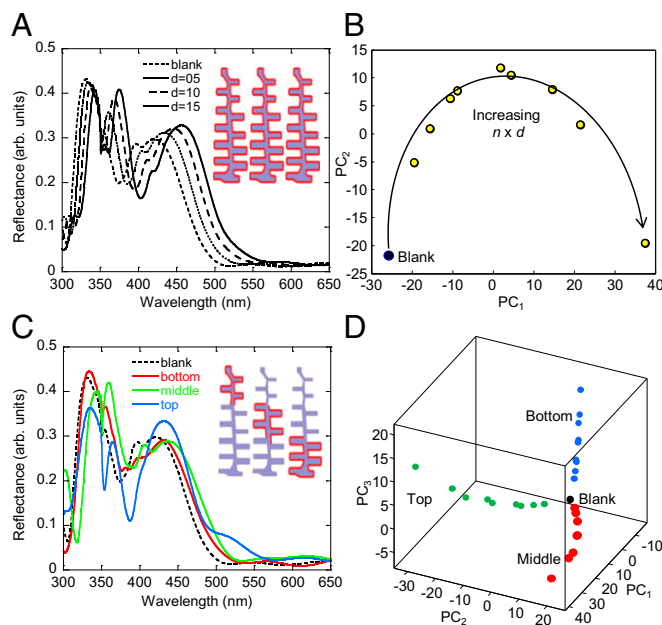


Fig. 4. Optical simulations of unusual vapor selectivity of *Morpho* butterfly scales using three vapors with $n = 1.3$, 1.4, and 1.5 and vapor concentrations related to condensed liquid layers of thickness $d = 5$, 10, and 15 nm. (A) Examples of reflectance spectra of uniform vapor adsorption onto the *Morpho* nanostructure. Vapor refractive index $n = 1.5$; $d = 5$, 10, and 15 nm. (B) PCA scores plot of spectral response of the *Morpho* nanostructure upon the uniform coverage $n \times d$ with adsorbed vapors. (C) Examples of reflectance spectra of “gradient” vapor adsorption onto three regions of the *Morpho* nanostructure covering only the top, middle, and bottom regions of the nanostructure, respectively. Vapor refractive index $n = 1.5$; $d = 15$ nm. (D) PCA scores plot of spectral response of the *Morpho* nanostructure upon the three-region coverage $n \times d$ with adsorbed vapors. Insets in A and C show the modeled distribution of vapors.

Reversed-Phase HPLC-Electrospray ToF-MS analysis. The chromatographic analysis was achieved using an Agilent 1200 Series HPLC system equipped with a 1200 Series photo-diode array detector in-line to an AB Sciex QStar Elite quadrupole time-of-flight mass spectrometer equipped with an electrospray interface. The separation was carried out using a Cadenza CW C18 1- $\times$ 50-mm reverse-phase column.

Optical Microscopy Characterization. Wing samples were stained in Nile red and DCM solutions in acetone for 10 min. Fluorescence and reflected light images were collected on an Olympus BX61 microscope using a 100 $\times$ chromatic aberration-corrected, air-operated, long-focal-length objective with the spatial resolution of 350 nm (according to the manufacturer's specification). Spectra were collected with an attached spectrograph (QE65000; Ocean Optics). The *Morpho* wings were scanned with a 40-nm step size, 100-nm $\cdot$ s $^{-1}$ stage scan rate.

Vapor Exposures. The vapor flow was arranged parallel to the ridges on the scales and delivered in the direction from the scale sockets to their distal edge. Different concentrations of vapors were delivered by 1/8-inch-diameter tubing at $\sim 45^\circ$ from the normal to the sample with a 450-mL/min total flow. Concentrations of generated analyte vapors were presented as P/P_0 , where P is vapor partial pressure and P_0 is the saturated vapor pressure. A hydration step between switching to different vapors was used to speed up the recovery of the wing sample to its baseline and was done by exposing the sample to water vapor at 0.35 P/P_0 . The hydration step was not needed for testing of porous silicon samples.

Reflectance Measurements and Data Analysis. A stabilized halogen light source (HL-2000-FHSA; Mikropack) and a fiber-optic spectrograph (QE65000; Ocean Optics) were used for reflectance measurements. The six-around-one

fiber-optic reflectance probe was positioned at $\sim 30^\circ$ from the normal to the sample surface. The size of the illuminated area of the sample was ~ 2 mm in diameter. PCA was performed upon autoscaling of spectra.

Optical Modeling. Finite element modeling was used to calculate the total light reflectance from *Morpho sulkowskyi*-like photonic structures. A unit cell containing a geometric representation of the butterfly's nanostructure was created from squared-off TEM images. Reflectance spectra from an infinite array of unit cells with a 700-nm ridge spacing were simulated for nanostructures with and without vapor overlayer conditions with the light incident normal to the butterfly scale and with electric field polarized parallel to the scale ridges.

Preparation of Porous Silicon as Control Sensing Material. Samples were prepared by electrochemical etching of boron-doped (100) oriented Si wafers with a resistivity $<0.002 \Omega\cdot\text{cm}^{-1}$. The etching solution was prepared by mixing three volumes of hydrofluoric acid [49% (wt/wt); Fisher Scientific] and one volume of ethanol [99.8% (wt/wt); Fisher Scientific]. Etching was done using an alternating current protocol (current density of 90 mA $\cdot$ cm $^{-2}$ for 1.5 s, followed by 380 mA $\cdot$ cm $^{-2}$ for 1.5 s). A total of ~ 30 cycles were finished before the porous Si surface was washed (ethanol and isopropanol) and dried (N_2).

ACKNOWLEDGMENTS. We thank H. Grade, M. Pietrzykowski, R. Schnoor, W. Shang, and V. Smentkowski (General Electric, GE), D. Gallagher and P. Lloyd (Air Force Research Laboratory), R. Bonam and J. Hartley (University at Albany), S. N. Gorb (University of Kiel), and V. Greanya and J. Abo-Shaeer (Defense Advanced Research Projects Agency, DARPA) for fruitful discussions. This work has been supported in part from DARPA Contract W911NF-10-C-0069 and from GE's Advanced Technology research funds.

- Mason CW (1926) Structural colors in insects. I. *J Phys Chem* 30:383–395.
- Ghiradella H, Aneshansley D, Eisner T, Silberglied RE, Hinton HE (1972) Ultraviolet reflection of a male butterfly: Interference color caused by thin-layer elaboration of wing scales. *Science* 178(4066):1214–1217.
- Vukusic P, Sambles JR (2003) Photonic structures in biology. *Nature* 424(6950):852–855.
- Kinoshita S, Yoshioka S, eds (2005) *Structural Colors in Biological Systems – Principles and Applications* (Osaka Univ Press, Osaka, Japan).
- Vukusic P, Sambles JR, Lawrence CR, Wootton RJ (1999) Quantified interference and diffraction in single *Morpho* butterfly scales. *Proc Biol Sci* 266(1427):1403–1411.
- Potyrailo RA, et al. (2007) *Morpho* butterfly wing scales demonstrate highly selective vapour response. *Nat Photonics* 1(2):123–128.
- Bielanski A, Deren J, Haber J (1957) Electric conductivity and catalytic activity of semiconducting oxide catalysts. *Nature* 179(4561):668–669.
- Janata J (2009) *Principles of Chemical Sensors* (Springer, New York).
- Persaud K, Dodd G (1982) Analysis of discrimination mechanisms in the mammalian olfactory system using a model nose. *Nature* 299(5881):352–355.
- Hierlemann A, Gutierrez-Osuna R (2008) Higher-order chemical sensing. *Chem Rev* 108(2):563–613.
- Potyrailo RA, Surman C, Nagraj NN, Burns A (2011) Materials and transducers toward selective wireless gas sensing. *Chem Rev* 111(11):7315–7354.
- Locke M (1966) The structure and formation of the cuticular layer in the epicuticle of an insect, *Calpodex ethlius* (Lepidoptera, Hesperidae). *J Morphol* 118(4):461–494.
- Pryor MGM (1940) On the hardening of the cuticle of insects. *Proc Biol Sci* 128(852):393–407.
- Pedrin N, Crespo R, Juárez MP (2007) Biochemistry of insect epicuticle degradation by entomopathogenic fungi. *Comp Biochem Physiol C Toxicol Pharmacol* 146(1–2):124–137.
- Neville AC (1975) *Biology of the Arthropod Cuticle* (Springer, Berlin).
- Greenspan P, Fowler SD (1985) Spectrofluorometric studies of the lipid probe, Nile red. *J Lipid Res* 26(7):781–789.
- Bechger L, Lodahl P, Vos WL (2005) Directional fluorescence spectra of laser dye in opal and inverse opal photonic crystals. *J Phys Chem B* 109(20):9980–9988.
- Zhou S, Cook KD (2000) Probing solvent fractionation in electrospray droplets with laser-induced fluorescence of a solvatochromic dye. *Anal Chem* 72(5):963–969.
- Sanchis MR, Calvo O, Fenollar O, García D, Balart R (2007) Surface modification of a polyurethane film by low pressure glow discharge oxygen plasma treatment. *J Appl Polym Sci* 105(3):1077–1085.
- Baklanov MR, Mogilnikov KP, Polovinkin VG, Dultsev FN (2000) Determination of pore size distribution in thin films by ellipsometric porosimetry. *J Vac Sci Technol B* 18(3):1385–1391.
- Rassart M, Simonis P, Bay A, Deparis O, Vigneron JP (2009) Scale coloration change following water absorption in the beetle *Hoplia coerules* (Coleoptera). *Phys Rev E Stat Nonlin Soft Matter Phys* 80(3 Pt 1):031910.
- Tamáskai I, et al. (2013) Color changes upon cooling of Lepidoptera scales containing photonic nanoarchitectures. *Key Eng Mater* 543:18–21.
- Li YY, et al. (2003) Polymer replicas of photonic porous silicon for sensing and drug delivery applications. *Science* 299(5615):2045–2047.
- Yang H, Jiang P (2011) Macroporous photonic crystal-based vapor detectors created by doctor blade coating. *Appl Phys Lett* 98:011104.
- Otto M (1999) *Chemometrics: Statistics and Computer Application in Analytical Chemistry* (Wiley-VCH, Weinheim, Germany).
- Wanekaya AK, Uematsu M, Breimer M, Sadik OA (2005) Multicomponent analysis of alcohol vapors using integrated gas chromatography with sensor arrays. *Sens Actuators B Chem* 110(1):41–48.
- Snow PA, Squire EK, Russell PSJ, Canham LT (1999) Vapor sensing using the optical properties of porous silicon Bragg mirrors. *J Appl Phys* 86(4):1781–1784.
- Burgess IB, et al. (2012) Wetting in color: Colorimetric differentiation of organic liquids with high selectivity. *ACS Nano* 6(2):1427–1437.
- Bjorklund RB, Zangooie S, Arwin H (1996) Color changes in thin porous silicon films caused by vapor exposure. *Appl Phys Lett* 69(20):3001–3003.
- Ruminski AM, Moore MM, Sailor MJ (2008) Humidity-compensating sensor for volatile organic compounds using stacked porous silicon photonic crystals. *Adv Funct Mater* 18(21):3418–3426.
- De Stefano L, et al. (2009) Hybrid polymer-porous silicon photonic crystals for optical sensing. *J Appl Phys* 106(2):023109.
- Yang H, Jiang P, Jiang B (2012) Vapor detection enabled by self-assembled colloidal photonic crystals. *J Colloid Interface Sci* 370(1):11–18.
- Zuccarello G, Scribner D, Sands R, Buckley LJ (2002) Materials for bio-inspired optics. *Adv Mater* 14(18):1261–1264.
- Claussen KU, Scheibel T, Schmidt H-W, Giesa R (2012) Polymer gradient materials: Can nature teach us new tricks? *Macromol Mater Eng* 297(10):938–957.
- Peisker H, Michels J, Gorb SN (2013) Evidence for a material gradient in the adhesive tarsal setae of the ladybird beetle *Coccinella septempunctata*. *Nature Comm* 4:1661.
- Ghiradella H (1994) Structure of butterfly scales: Patterning in an insect cuticle. *Microsc Res Tech* 27(5):429–438.
- Kennaway R, Coen E, Green A, Bangham A (2011) Generation of diverse biological forms through combinatorial interactions between tissue polarity and growth. *PLoS Comput Biol* 7(6):e1002071.
- Huber G, et al. (2005) Evidence for capillarity contributions to gecko adhesion from single spatula nanomechanical measurements. *Proc Natl Acad Sci USA* 102(45):16293–16296.
- Voigt D, Peisker H, Gorb SN (2009) Visualization of epicuticular grease on the covering wings on the Colorado Potato Beetle: A scanning probe approach. In *Applied Scanning Probe Methods XIII*, eds Bhushan B, Fuchs H (Springer, Heidelberg), pp 1–16.
- Lee SY, et al. (2010) Spatial and spectral detection of protein monolayers with deterministic aperiodic arrays of metal nanoparticles. *Proc Natl Acad Sci USA* 107(27):12086–12090.
- Kolle M, et al. (2010) Mimicking the colourful wing scale structure of the *Papilio blumei* butterfly. *Nat Nanotechnol* 5(7):511–515.
- Watanabe K, et al. (2005) Brilliant blue observation from a *Morpho*-butterfly-scale quasi-structure. *Jpn J Appl Phys* 44(1):L48–L50.
- Aryal M, et al. (2012) Large area nanofabrication of butterfly wing's three dimensional ultrastructures. *J Vac Sci Technol B* 30(6):061802.